

Step-by-Step Execution & Deployment Guide

Real-Time Twitter Sentiment Analyzer

Github: <https://github.com/Yash-byte-del/Data-Analytics-ELC-Activity-5th-Sem-/tree/main>

Streamlit: <https://tweetsentimeter.streamlit.app/>

Dataset: [bdstar/twitter-sentiment-analysis · Datasets at Hugging Face](#)

Project Folder Contents

1. notebook.ipynb – Model training and analysis
2. app.py – Streamlit web application
3. sentiment_model.pkl – Trained sentiment model
4. tfidf_vectorizer.pkl – TF-IDF feature extractor
5. requirements.txt – Required Python libraries
6. Execution_and_Deployment_Guide – Deployment steps and important links

Step 1: Environment Setup

Install Python 3.9 or above. Ensure Jupyter Notebook is installed. Run the following command inside the project folder to install dependencies:

pip install -r requirements.txt

Step 2: Running notebook.ipynb

Open notebook.ipynb in Jupyter Notebook and execute all cells from top to bottom. The notebook performs dataset loading, preprocessing, exploratory data analysis, feature extraction using TF-IDF, model training, evaluation, and saving of model files.

Step 3: Understanding app.py

The app.py file loads the trained model and vectorizer, accepts live user input, processes the text, predicts sentiment, and displays results in real time using Streamlit.

Step 4: Run the Application Locally

From the project directory, run:

streamlit run app.py

A browser window will open showing the application.

Step 5: Real-Time Simulation

The system simulates real-time sentiment analysis by processing live user input and generating instant predictions with low latency.

Step 6: Deployment Summary

Upload all files to GitHub. Log in to Streamlit Cloud, select the repository, choose app.py as the main file, and deploy